

Lemma T: interior concavity, the domain-free case, and what separability is for

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Abstract

Let Φ be concave and 1-Lipschitz on an integer interval, Ψ convex and 1-Lipschitz on an integer interval, and $G(s) = \sum_{i+j=s} (\Phi(i) - \Psi(j) + 1)_+$. Lemma T asserts that G is log-concave; it is the whole of condition (A) at $m = 2$ for cylindric Kostka numbers, via the exact reduction [1, Prop. 4.3]. We prove three things. First, a *local* theorem: at every antidiagonal whose cell-support is strictly interior to the box, G satisfies the *concavity* inequality $G(s-1) + G(s+1) \leq 2G(s)$ — strictly stronger than log-concavity, and proved in half a page. Second, the resulting complete proof of Lemma T in the *domain-free* case, where Φ and Ψ are defined on all of $\mathbb{Z}$; this is exactly the functional form that the cylindric reduction produces, minus the two window constraints. Third, a converse localisation: over 2 781 720 antidiagonals, concavity fails at 846 720 of them and *not one* of these satisfies the interiority hypothesis, so the hypothesis is not an artifact — it is the precise boundary of the concavity method. We also record two refutations that fix the scope of any future proof: the non-separable analogue is false, and so is the general “king-fan” geometric reformulation. Lemma T in full remains open; the gap is the two window constraints and nothing else.

1 Statement, and a cosmetic simplification

Throughout, $(x)_+ = \max(x, 0)$, and PF_2 means: nonnegative, support an interval of $\mathbb{Z}$, and $g(s)^2 \geq g(s-1)g(s+1)$ for all s .

Conjecture 1 (Lemma T). Let Φ be concave and 1-Lipschitz on $[A, B] \subset \mathbb{Z}$ and Ψ convex and 1-Lipschitz on $[C, D] \subset \mathbb{Z}$ (both extended by $\mp\infty$). Then

$$G(s) = \sum_{\substack{i+j=s \\ i \in [A, B], j \in [C, D]}} (\Phi(i) - \Psi(j) + 1)_+ \tag{1}$$

is log-concave.

The asymmetry between “concave” and “convex” in the statement is cosmetic. If $\Theta := -\Psi$ then Θ is concave and 1-Lipschitz, and $\Phi(i) - \Psi(j) + 1 = \Phi(i) + \Theta(j) + 1$, so (1) reads

$$G(s) = \sum_{i+j=s} (F(i, j))_+, \quad F(i, j) = \Phi(i) + \Theta(j) + 1, \tag{2}$$

with Φ, Θ *both* concave and 1-Lipschitz. So F is a separable concave function on a box with both partial differences in $\{-1, 0, 1\}$, and G is the antidiagonal marginal of F_+ . We use whichever form is convenient.

Two normalisations, used without comment below. An integer-valued concave 1-Lipschitz function on $[A, B]$ has non-increasing increments in $\{1, 0, -1\}$, hence increment word $1^a 0^b (-1)^c$ with $a + b + c = B - A$; and G depends on the data only through $A + C$ and $\Phi(A) - \Psi(C)$, so one may take $A = C = 0$ and $\Psi(C) = 0$. The whole parameter space is therefore $(L_1, a_1, c_1; L_2, a_2, c_2; d)$ with a single integer shift d . (Verified against 20 000 profiles generated *without* this parametrisation: no mismatch.)

2 Notation for one antidiagonal

Fix s . Write

$$\varphi'(i) = \Phi(i) - \Phi(i-1) \in \{-1, 0, 1\}, \quad \psi'(x) = \Psi(x) - \Psi(x-1) \in \{-1, 0, 1\},$$

so φ' is non-increasing (Φ concave) and ψ' is non-decreasing (Ψ convex); both lie in $\{-1, 0, 1\}$ by 1-Lipschitz. Put $J = [A, B] \cap [s-D, s-C]$ and, for $i \in J$,

$$v_i = \Phi(i) - \Psi(s-i) + 1,$$

the *untruncated* value of the cell $(i, s-i)$. Let

$$I = \{i \in J : v_i \geq 1\},$$

the set of i contributing to $G(s)$, and assume $I \neq \emptyset$. Since Φ is concave and $i \mapsto -\Psi(s-i)$ is concave, v is concave on J ; hence $I = [i_0, i_1]$ is an interval and

$$V := \sum_{i \in I} v_i = G(s).$$

Write $[x_0, x_1] = [s-i_1, s-i_0]$ for the corresponding range of $j = s-i$, and

$$l := \psi'(x_0), \quad r := \psi'(x_1+1).$$

Finally, for $i \in I$ set $\delta_i = \psi'(s-i)$ and $\varepsilon_i = \psi'(s+1-i)$, so that $\varepsilon_i \geq \delta_i$ and the cells of i on the neighbouring antidiagonals are

$$\Phi(i) - \Psi(s-1-i) + 1 = v_i + \delta_i, \quad \Phi(i) - \Psi(s+1-i) + 1 = v_i - \varepsilon_i. \quad (3)$$

This is the one computation that organises everything: *at fixed i the three cells on antidiagonals $s-1, s, s+1$ differ only through Ψ* , so a single concave sequence v and the non-decreasing increment sequence ψ' control all three.

Definition 2. Antidiagonal s is *interior*, written (INT), if

$$i_0 - 1 \geq A, \quad i_1 + 1 \leq B, \quad x_0 - 1 \geq C, \quad x_1 + 1 \leq D,$$

i.e. if the cell-support I is strictly inside the box in both coordinates.

3 The support is always an interval

Proposition 3. *supp G is an interval of $\mathbb{Z}$.*

Proof. Let $\tilde{\Phi}, \tilde{\Theta}$ be the piecewise-linear interpolations of Φ, Θ , extended by $-\infty$ off $[A, B], [C, D]$; they are concave on $\mathbb{R}$ with breakpoints at integers. Suppose $G(s') > 0$ and $G(s'') > 0$ with $s' < s < s''$. Choose lattice points $(i', j'), (i'', j'')$ in the box with $i' + j' = s', i'' + j'' = s''$ and $F > 0$ at both. The segment joining them meets the line $i + j = s$ at a point (α, β) with $\tilde{\Phi}(\alpha) + \tilde{\Theta}(\beta) + 1 > 0$, by concavity of $\tilde{\Phi}(\cdot) + \tilde{\Theta}(\cdot) + 1$ along the segment. Now $h(\xi) := \tilde{\Phi}(\xi) + \tilde{\Theta}(s - \xi) + 1$ is concave and piecewise linear in ξ with breakpoints at integers, so its maximum over $\mathbb{R}$ is attained at an integer i ; and $h(i) \geq h(\alpha) > 0$. Since $h = -\infty$ off the box, $i \in [A, B]$ and $s - i \in [C, D]$. Hence $G(s) \geq h(i) > 0$. $\square$

Remark 4. Proposition 3 uses separability (through h having integer breakpoints) and needs no Lipschitz hypothesis. It already fails for non-separable concave F : see Remark 17.

4 The interior concavity theorem

Everything in this section is local at one s .

Lemma 5 (two-sided decay). *If $i \in J$ and $i \geq i_1 + 2$ or $i \leq i_0 - 2$, then $v_i \leq -1$.*

Proof. $v_{i_1} \geq 1$ and $v_{i_1+1} \leq 0$, so the increment $v_{i_1+1} - v_{i_1} \leq -1$. As v is concave its increments are non-increasing, so every later increment is ≤ -1 and $v_{i_1+2} \leq v_{i_1+1} - 1 \leq -1$, and v decreases thereafter. Symmetrically on the left. $\square$

Lemma 6 (the sign of the two end slopes). *Assume (INT). Then $l \leq 0 \leq r$.*

Proof. By (INT) the cell $(i_0 - 1, x_1 + 1)$ lies in the box, so v_{i_0-1} is defined and $v_{i_0-1} \leq 0 < 1 \leq v_{i_0}$. Now

$$v_{i_0} - v_{i_0-1} = [\Phi(i_0) - \Phi(i_0 - 1)] + [\Psi(x_1 + 1) - \Psi(x_1)] = \varphi'(i_0) + r,$$

and the left side is ≥ 1 . Since $\varphi'(i_0) \leq 1$ we get $r \geq 1 - \varphi'(i_0) \geq 0$. Likewise $(i_1 + 1, x_0 - 1)$ lies in the box, $v_{i_1+1} \leq 0 < 1 \leq v_{i_1}$, and

$$v_{i_1+1} - v_{i_1} = \varphi'(i_1 + 1) + [\Psi(x_0) - \Psi(x_0 - 1)] = \varphi'(i_1 + 1) + l \leq -1,$$

so $l \leq -1 - \varphi'(i_1 + 1) \leq 0$ because $\varphi'(i_1 + 1) \geq -1$. $\square$

Lemma 6 is where Φ 's Lipschitz bound is spent, and it is spent *only* here.

Remark 7 (machine-checked). Lemmas 5, 6 and 9 are formalised in `lean/tworow_d4_kernel/TwoRowD4Kernel/` sorry-free, on the standard three axioms (`propext`, `Classical.choice`, `Quot.sound`): `decay_left/decay_right` for Lemma 5, `LemmaT.Setup.lem_signs` for Lemma 6, and `LemmaT.Setup.lem_offJ_lower/lem_offJ_upper` for Lemma 9. The intermediate inequalities $r \geq 1 - \varphi'(i_0)$ and $l \leq -1 - \varphi'(i_1 + 1)$ of the proof below are `key_r` and `key_l`; Lemma 9 cites *those*, not the weaker published conclusion $l \leq 0 \leq r$.

Every hypothesis is restricted to the box in the Lean statement, so that the step “the cell lies in the box, so v_i is *defined*” is a real obligation rather than a no-op: (INT) is used twice, once to put $i_0 - 1$ and $i_1 + 1$ inside the box so the cell-support description applies there, and once to put i_0 and $i_1 + 1$ inside the range where Φ 's Lipschitz bound is available. Three negative controls

accompany the formalisation. **Control.wit** is an explicit instance ($A = 0, B = 5, C = 0, D = 2, s = 4, I = \{3\}, \Phi \equiv 0, \Psi = |x - 1|$) satisfying (INT) in which *both* off- J cells lie in the box, so none of the four conclusions is vacuous; it also shows the bound of Lemma 9 is tight, both cells having value exactly 0. **Control.noLo** violates the single (INT) clause $i_0 - 1 \geq A$ — the other three hold, and $x_1 + 1 \leq D$, so r is still an increment of Ψ inside its domain — and has $r = -1$, so Lemma 6 fails: that clause is load-bearing. **Control. Φ _lipschitz_needed** drops only Φ 's Lipschitz bound, keeping (INT) in full, and again produces $r = -1$: with $\Phi = (-10, 0, 0, \dots)$ concave on $[0, 6]$, $\Psi(x) = -x, s = 4, I = [1, 4]$. In both controls the other half $l \leq 0$ still holds, so the controls say which hypothesis feeds which inequality.

Lemma 8 (no truncation inside I). *Assume (INT). For every $i \in I$ the cells $(i, s - 1 - i)$ and $(i, s + 1 - i)$ lie in the box, and their values are exactly $v_i + \delta_i \geq 0$ and $v_i - \varepsilon_i \geq 0$.*

Proof. For $i \in I$ we have $s - 1 - i \geq s - 1 - i_1 = x_0 - 1 \geq C$ and $s - 1 - i \leq x_1 - 1 \leq D$; similarly $s + 1 - i \leq x_1 + 1 \leq D$ and $s + 1 - i \geq x_0 + 1 \geq C$. So both cells are in the box and (3) applies. Finally $v_i \geq 1$ while $|\delta_i| \leq 1$ and $|\varepsilon_i| \leq 1$, whence $v_i + \delta_i \geq 0$ and $v_i - \varepsilon_i \geq 0$. $\square$

This is what the Lipschitz bound on Ψ buys: *the positive part in (1) is inactive on the whole cell-support*. All truncation lives outside I , and by Lemma 5 that is two cells.

Lemma 9 (boundary contributions). *Assume (INT) and set*

$$E_- = \sum_{i \in J \setminus I} (\Phi(i) - \Psi(s - 1 - i) + 1)_+, \quad E_+ = \sum_{i \in J \setminus I} (\Phi(i) - \Psi(s + 1 - i) + 1)_+,$$

each term taken to be 0 when its cell is outside the box. Then $E_- \leq r$ and $E_+ \leq -l$.

Proof. By Lemma 5 only $i = i_0 - 1$ and $i = i_1 + 1$ can contribute, since for the other $i \in J \setminus I$ we have $v_i \leq -1$ and hence $(v_i + \delta_i)_+ \leq (-1 + 1)_+ = 0$ and $(v_i - \varepsilon_i)_+ \leq (-1 + 1)_+ = 0$. There are four cells to bound, and ψ' is non-decreasing throughout.

E_- , at $i = i_0 - 1$: the cell is $(i_0 - 1, x_1)$, of value $v_{i_0-1} + \psi'(x_1 + 1) = v_{i_0-1} + r \leq 0 + r = r$.

E_- , at $i = i_1 + 1$: the cell is $(i_1 + 1, x_0 - 2)$, of value $v_{i_1+1} + \psi'(x_0 - 1) \leq 0 + l \leq 0$, using $\psi'(x_0 - 1) \leq \psi'(x_0) = l \leq 0$ (Lemma 6). So it contributes 0.

E_+ , at $i = i_1 + 1$: the cell is $(i_1 + 1, x_0)$, of value $v_{i_1+1} - \psi'(x_0) = v_{i_1+1} - l \leq -l$.

E_+ , at $i = i_0 - 1$: the cell is $(i_0 - 1, x_1 + 2)$, of value $v_{i_0-1} - \psi'(x_1 + 2) \leq 0 - r \leq 0$, using $\psi'(x_1 + 2) \geq \psi'(x_1 + 1) = r \geq 0$. So it contributes 0.

(Any of these cells lying outside the box contributes 0, which only strengthens the bounds.)

Adding up gives the claim. $\square$

Lemma 10 (the two off- J cells vanish). *Assume (INT). The antidiagonals $s \pm 1$ each meet one cell whose i lies outside J , namely $i = s - 1 - D$ for $G(s - 1)$ (at $j = D$) and $i = s + 1 - C$ for $G(s + 1)$ (at $j = C$). Both cells have value 0.*

Proof. The index sets are $J = [A, B] \cap [s - D, s - C]$ and, for $s - 1$, $[A, B] \cap [s - 1 - D, s - 1 - C]$; they differ only at the two ends, so $i = s - 1 - D$ is the sole index contributing to $G(s - 1)$ with $i \notin J$. If $i \notin [A, B]$ the cell is absent, so assume $i \in [A, B]$ and put $i' = i + 1 = s - D$. From (INT) we have $x_1 + 1 \leq D$, i.e. $i_0 \geq s + 1 - D = i + 2$, so $i' \leq i_0 - 1$; also $s - i' = D \in [C, D]$ and $A \leq i' \leq i_0 - 1 \leq B$, so $i' \in J$ and $v_{i'}$ is defined. Since $\Phi(i) = \Phi(i') - \varphi'(i')$,

$$\Phi(i) - \Psi(D) + 1 = v_{i'} - \varphi'(i').$$

If $i' \leq i_0 - 2$ then $v_{i'} \leq -1$ by Lemma 5 and $\varphi'(i') \geq -1$, so the value is ≤ 0 . If $i' = i_0 - 1$ then $v_{i'} \leq 0$, so the value is $\leq -\varphi'(i_0 - 1)$, which is ≤ 0 unless $\varphi'(i_0 - 1) = -1$; but φ' is non-increasing, so then $\varphi'(i_0) \leq -1$, and Lemma 6 would force $r \geq 1 - \varphi'(i_0) \geq 2$, impossible since $r = \psi'(x_1 + 1) \in \{-1, 0, 1\}$. So that case does not occur.

For $G(s + 1)$ and $i = s + 1 - C$ put $i' = i - 1 = s - C$. From (INT), $x_0 - 1 \geq C$ gives $i_1 \leq s - 1 - C = i - 2$, so $i' \geq i_1 + 1$, and as before $i' \in J$. Here $\Phi(i) = \Phi(i') + \varphi'(i)$, so the value is $v_{i'} + \varphi'(i)$. If $i' \geq i_1 + 2$ then $v_{i'} \leq -1$ and $\varphi'(i) \leq 1$, giving ≤ 0 . If $i' = i_1 + 1$ then $v_{i'} \leq 0$ and the value is $\leq \varphi'(i_1 + 2)$, which is ≤ 0 unless $\varphi'(i_1 + 2) = 1$; then $\varphi'(i_1 + 1) \geq 1$, and Lemma 6 would force $l \leq -1 - \varphi'(i_1 + 1) = -2$, impossible. $\square$

Lemma 11 (telescoping). $\sum_{i \in I} \varepsilon_i - \sum_{i \in I} \delta_i = r - l$.

Proof. As i runs over $I = [i_0, i_1]$, $s - i$ runs over $[x_0, x_1]$ and $s + 1 - i$ over $[x_0 + 1, x_1 + 1]$. Hence $\sum_I \delta_i = \sum_{x=x_0}^{x_1} \psi'(x) = \Psi(x_1) - \Psi(x_0 - 1)$ and $\sum_I \varepsilon_i = \Psi(x_1 + 1) - \Psi(x_0)$, so the difference is $[\Psi(x_1 + 1) - \Psi(x_1)] - [\Psi(x_0) - \Psi(x_0 - 1)] = \psi'(x_1 + 1) - \psi'(x_0) = r - l$. $\square$

Theorem 12 (interior concavity). *Let Φ be concave and 1-Lipschitz on $[A, B]$, Ψ convex and 1-Lipschitz on $[C, D]$, and let G be as in (1). If $G(s) > 0$ and antidiagonal s is interior in the sense of Definition 2, then*

$$G(s - 1) + G(s + 1) \leq 2G(s).$$

Proof. Put $D_\Sigma = \sum_{i \in I} \delta_i$ and $E_\Sigma = \sum_{i \in I} \varepsilon_i$. By Lemma 7 every $i \in I$ contributes exactly $v_i + \delta_i$ to $G(s - 1)$ and exactly $v_i - \varepsilon_i$ to $G(s + 1)$, while $J \setminus I$ contributes E_- and E_+ ; and the two indices outside J contribute nothing, by Lemma 9. Hence

$$G(s - 1) = V + D_\Sigma + E_-, \quad G(s + 1) = V - E_\Sigma + E_+.$$

Therefore, using Lemma 10 and then Lemma 8,

$$G(s - 1) + G(s + 1) - 2G(s) = -(E_\Sigma - D_\Sigma) + E_- + E_+ = -(r - l) + E_- + E_+ \leq -(r - l) + r + (-l) = 0. \quad \square$$

The proof is a two-line cancellation once the four lemmas are in place, and it is worth saying exactly where each hypothesis went: Ψ convex gives ψ' non-decreasing (Lemmas 8, 10); Ψ 1-Lipschitz gives Lemma 7; Φ concave gives concavity of v (Lemma 5); Φ 1-Lipschitz gives Lemma 6. Every hypothesis is used, once.

Corollary 13. *If antidiagonal s is interior for every $s \in \text{supp } G$, then G is concave on its support, and $G \in \text{PF}_2$.*

Proof. Theorem 11 gives concavity on the interior of the support; Proposition 3 gives that the support is an interval; now [1, Lem. 2.1] (concave + nonnegative + interval support $\Rightarrow \text{PF}_2$, by AM–GM) applies. $\square$

5 The domain-free case

Corollary 14 (Lemma T, domain-free). *Let $\Phi : \mathbb{Z} \rightarrow \mathbb{Z}$ be concave and 1-Lipschitz with $\Phi(i) \rightarrow -\infty$ in both directions, and $\Psi : \mathbb{Z} \rightarrow \mathbb{Z}$ convex and 1-Lipschitz with $\Psi(j) \rightarrow +\infty$ in both directions. Equivalently,*

$$\Phi(i) = \min(i - p, M, q - i), \quad \Psi(j) = \max(u - j, m, j - v)$$

for integers $p \leq q$, M , u , v , m . Then $G(s) = \sum_{i+j=s} (\Phi(i) - \Psi(j) + 1)_+$ is concave on its support, and in particular log-concave. So Lemma T holds in this case.

Proof. The stated form is forced: an integer concave 1-Lipschitz Φ on $\mathbb{Z}$ has non-increasing increments in $\{-1, 0, 1\}$, and $\Phi \rightarrow -\infty$ both ways forces the increment word $1^\infty 0^b (-1)^\infty$, i.e. a tent with a plateau at height M ; dually for Ψ . With $A = -\infty$, $B = +\infty$, $C = -\infty$, $D = +\infty$ the four conditions of Definition 2 hold vacuously, so every antidiagonal with $G(s) > 0$ is interior. Now apply Corollary 12, noting that G has finite support because $v_i \rightarrow -\infty$. $\square$

This is not a vacuous specialisation. The reduction [1, Prop. 4.3] produces exactly

$$\Phi(i) = \min(i, u - 1 - i, T_+), \quad \Psi(j) = \max(u - j, j - n + 1, T_-),$$

which are a tent-with-cap and a V-with-floor — precisely the shapes of Corollary 13. The *only* difference between Corollary 13 and Lemma T is the pair of window constraints $i \in [A, B]$, $j \in [C, D]$.

6 The gap, and why it is exactly the windows

Corollary 12 is conditional on interiority, and the condition bites: over the exhaustive family of §1 with profile lengths ≤ 8 there are 2 781 720 antidiagonals with $G(s) > 0$, of which 52 064 are interior. It is therefore essential to know whether the hypothesis is an artifact of the proof or a real boundary. It is real:

Proposition 15 (converse localisation, computational). *Over the same 2 781 720 antidiagonals, the concavity inequality $G(s-1) + G(s+1) \leq 2G(s)$ fails at 846 720 of them, and none of these is interior. Moreover, over all interior antidiagonals, each of $l \leq 0 \leq r$ (Lemma 6), $E_- \leq r$ and $E_+ \leq -l$ (Lemma 8), and the concavity conclusion itself, holds with zero exceptions.*

Coverage should be stated plainly: on this same family (profile lengths ≤ 7 , 112 896 instances) (INT) holds at *every* $s \in \text{supp } G$ in only 1326 instances (1.2%), and at at least one s in 8078. Corollary 12 is therefore a thin criterion on *short* profiles, where the box is small and the cell-support almost always abuts it; its force is in the limit Corollary 13, where the box is absent altogether.

So (INT) is exactly the frontier of the concavity method, and the residual difficulty is entirely multiplicative: at a non-interior s , concavity genuinely fails and log-concavity must be obtained from the exact defect. That defect is available in closed form. With $\alpha = G(s-1) - G(s)$ and $\beta = G(s+1) - G(s)$ one has identically $G(s)^2 - G(s-1)G(s+1) = -G(s)(\alpha + \beta) - \alpha\beta$, and in the no-truncation regime Lemmas 10–8 give the clean identity

$$G(s)^2 - G(s-1)G(s+1) = V(r-l) + D_\Sigma E_\Sigma, \quad (4)$$

where $D_\Sigma E_\Sigma < 0$ forces $r-l = 2$ and $|D_\Sigma| = 1$, so the right side is $\geq 2V-1 > 0$. (Formula (4) and its boundary-corrected version were checked against the definition on all 2 781 720 antidiagonals: no mismatch.)

Question 16. At a non-interior s , (INT) fails because I abuts $\partial([A, B] \times [C, D])$. Then $r < 0$ or $l > 0$ becomes possible and Lemma 6 is lost — but the lost cell v_{i_0-1} or v_{i_1+1} is also *absent*, which removes a term from $E_\mp$. Is there a bookkeeping in which the failure of Lemma 6 is always paid for by the simultaneous loss in Lemma 8?

That is the whole of what remains. We were not able to close it; what follows fences off three routes that cannot.

7 Scope: three refutations

Remark 17 (the birth range). The evidence for Lemma T in [1, Rem. 4.5] was 0 failures in an exhaustive enumeration of profiles of length ≤ 6 with base shifts in $[-4, 6]$ (29 000 instances). A perfect score on the set that produced a conjecture is not evidence of independence. We therefore re-tested first: exhaustively over *all* profiles of length ≤ 14 with all shifts, 15 052 800 instances, 0 failures; randomly at lengths 20–80, 616 054 instances, 0 failures; and in the degenerate corners (one profile a single point against one of length up to 120; both profiles of length ≤ 3), 372 710 instances, 0 failures. Lengths L_1, L_2 were varied independently throughout. Lemma T is not an artifact of its discovery range.

Remark 18 (separability is load-bearing). Form (2) invites dropping separability: if F is concave on a box with both partial differences in $\{-1, 0, 1\}$, is $s \mapsto \sum_{i+j=s} F(i, j)_+$ log-concave? No. Among 200 000 random such F (generated as minima of affine functions with coefficients in $\{-1, 0, 1\}$, of which 38 883 were genuinely non-separable) there are 1679 failures. The smallest witness is

$$F = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \min(-i + j + 1, i - j + 1) \quad \text{on } [0, 1]^2, \quad G = (1, 0, 1),$$

which fails by an *internal zero*: separability is needed even for Proposition 3. The smallest witness with interval support is

$$F(i, j) = \min(i - j + 3, 1) \quad \text{on } [0, 6] \times [0, 4], \quad G = (1, 2, 3, 3, 4, 4, 5, 4, 3, 2, 1),$$

failing at $s = 3$ ($9 < 12$) and $s = 5$ ($16 < 20$). What this witness *incidentally* satisfies is the informative part: F_+ is log-concave along every lattice line (rows, columns, and both diagonals), and its mixed second difference is constant $= 2$ (a separable F has mixed second difference $\equiv 0$). Hence *no pointwise log-concavity hypothesis on the summand can imply Lemma T*: any proof must use the separable form $(\Phi(i) + \Theta(j) + 1)_+$, not merely concavity of F or log-concavity of F_+ .

Remark 19 (the geometric reformulation, and why the king fan is not enough). Substituting $j = s - i$ and counting the integer t with $\Psi(j) \leq t \leq \Phi(i)$,

$$G(s) = \#\{(i, t) \in \mathbb{Z}^2 : i \in [A, B], s - i \in [C, D], \Psi(s - i) \leq t \leq \Phi(i)\}.$$

Using $A_t = \{i : \Phi(i) \geq t\} = [A + (t - \Phi(A))_+, B - (t - \Phi(B))_+]$ — the level sets of a concave 1-Lipschitz integer function move at speed exactly 1, with no jumps — all the constraints bound only i , t , $i - t$ and $i + t$, and exactly four of them move with s , each by $+1$. Hence

$$G(s) = \#(P \cap (Q + se_1) \cap \mathbb{Z}^2)$$

for fixed convex P, Q whose edge normals lie in the *king fan* $\{\pm e_1, \pm e_2, \pm(e_1 - e_2), \pm(e_1 + e_2)\}$: a convolution of two lattice-polygon indicators along a line, with the 1-Lipschitz hypothesis converted into a fan condition. This reformulation is *false* in that generality: among 2226 random king-fan pairs there are 3 failures, e.g. $G = (0, 0, 3, 2, 2, 1, 1)$ with $9 < 18$ at $s = 3$. Every witness has a *degenerate* polygon, pinned to a diagonal line ($\text{lo} = \text{hi}$ for $i - t$). In our situation P bounds $i - t$ only from below and $i + t$ only from above, while Q does the reverse, so neither can degenerate that way. The complementarity of P and Q is therefore load-bearing and is not implied by the fan condition.

Remark 20 (decompose-and-pair cannot work). The symmetrisation identity behind [1, §5] reduces $G = \sum_R G_R$ to $G_R \in \text{PF}_2$ for each R plus the pair inequalities $2G_R(s)G_{R'}(s) \geq G_R(s-1)G_{R'}(s+1) + G_R(s+1)G_{R'}(s-1)$. This condition is sufficient but far from necessary, and for the natural decompositions it simply fails. We tested three on 176 400 instances: peeling the last column ($G = G' + h$, $h(\sigma) = (\Phi(\sigma - D) + \Theta(D) + 1)_+$, with G' again a Lemma T instance, so that the pair inequality would give an induction on $D - C$ with a one-column base case); peeling the last row; and splitting at the minimum of Ψ . Each fails on 96 837, 96 837 and 67 996 instances respectively. The mechanism is visible in the smallest witness: G' supported on $\{0, 1\}$ and h on $\{2\}$ give $2G'(1)h(1) = 0 < 1 = G'(0)h(2)$, while $G = G' + h = (1, 1, 1)$ is perfectly log-concave. Whenever the summands have different supports the pair inequality is strictly stronger than what is wanted. So no decomposition of G into pieces with unequal supports can be completed by pairwise comparison, and in particular column- or row-induction is closed.

8 Computational verification

All computations are in `scratch/lemmaT/`.

1. **Harness control.** The two negative data points of [1, Rem. 4.5] were reproduced exactly: $\Phi = (-1, 0, 1)$, $\Psi = (-3, 0, 4)$ gives $G = (3, 4, 6, 2)$ with $16 < 18$, and $\Phi = (3, 5, 5)$, $\Psi = (3, 1, 2, 3, 4)$ gives $G = (1, 6, 10, 10, 7, 5, 2)$ with $49 < 50$. (Our G carries one further trailing zero in the first case; the violation is identical.)
2. **Parametrisation control.** 20 000 concave/convex 1-Lipschitz profile pairs were generated by rejection *without* the increment-word normal form; the normal form, the invariance of G under a common value shift, and the reduction to a single shift d each held with 0 violations.
3. **Lemma T.** Remark 16: 16 041 564 instances, 0 failures.
4. **Theorem 11** and its three ingredient lemmas, and Proposition 14: 2 781 720 antidiagonals, 0 exceptions in each of four separate predicted quantities, and 0 of the 846 720 concavity failures interior.
5. **Lemma 9:** over the 52 064 interior antidiagonals the off- J cell exists inside $[A, B]$ on 19 403 occasions and is zero on every one of them. This lemma was missing from the first draft of the proof, where the step was asserted with a reference to Lemma 5 — which cannot apply, since v_i is not even defined off J .
6. **Corollary 13:** 12 058 domain-free instances (tent-with-cap against V-with-floor, over all of $\mathbb{Z}$); support an interval in all, G concave on its support in all, 0 log-concavity failures.
7. **The case analysis, enumerated.** An independent implementation over 1 403 305 domain-free antidiagonals, asserting every intermediate quantity inline ($|\delta_i|, |\varepsilon_i| \leq 1$; $\varepsilon_i \geq \delta_i$; no truncation inside I ; $E_\Sigma - D_\Sigma = r - l$), ran to completion with no assertion failure, no $G(s)^2 < G(s-1)G(s+1)$, and $E_\pm \in \{0, 1\}$. Exactly nine (g, E_-, E_+, l, r) cases occur, and

they are exactly the nine the lemmas permit:

g	E_-	E_+	l	r	g	E_-	E_+	l	r
0	0	0	0	0	2	0	0	-1	1
1	0	0	-1	0	2	0	1	-1	1
1	0	0	0	1	2	1	0	-1	1
1	0	1	-1	0	2	1	1	-1	1
1	1	0	0	1					

Note what is absent: l never exceeds 0 and r is never negative (Lemma 6); $E_- = 1$ occurs only in rows with $r = 1$ and $E_+ = 1$ only in rows with $l = -1$ (Lemma 8); and $g = r - l \geq E_- + E_+$ in every row, which is Theorem 11.

8. **Identity** (4): verified against the definition, 0 mismatches on 2 781 720 antidiagonals (166 035 of them in the no-truncation regime).

A methodological note, since it cost us a false step. Our first test of “ F is concave on $\mathbb{Z}^2$ ” asserted the diagonal inequalities $F(i, j) + F(i+1, j+1) \leq F(i+1, j) + F(i, j+1)$. These do *not* follow from concavity: both pairs have midpoint $(i + \frac{1}{2}, j + \frac{1}{2})$, so concavity bounds each sum by $2F(i + \frac{1}{2}, j + \frac{1}{2})$ and compares neither. The error surfaced only because the generator was concave by construction and was refuted by its own test — the control caught it, not the reasoning.

9 Status

- **Proved.** Proposition 3 (supp G is an interval); Theorem 11 (interior concavity); Corollary 12; Corollary 13 (Lemma T domain-free, with concavity). Together with [1, Prop. 4.3] this closes condition (A) at $m = 2$ for every slice whose two window constraints are inactive on the cell-support.
- **Open.** Lemma T in full, i.e. Conjecture 1. The gap is precisely the non-interior antidiagonals, where Lemma 6 fails. Nothing else in the argument is conditional.
- **Closed off.** Dropping separability (Remark 17); the general king-fan geometric statement; decomposition-plus-pairwise-comparison in any form with unequal supports (Remark 19), including row/column induction.

References

- [1] Clio, *Log-concavity of cylindric Kostka numbers: condition (A) at $m = 2$, proofs/2026-09-30-c1-cylindric-kostka-logconcavity.tex*, 30 September 2026. (Referenced here: Lem. 2.1 `lem:conc`, Lem. 2.2 `lem:trunc`, Prop. 4.3 `prop:m2red`, Rem. 4.5 `rem:sharp`, Thm. 5.4 `thm:regions`, §5 `sec:gap`.)